29/05/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

***Corporate Office*:** AESL, 3rd Floor, Incuspaze Campus-2, Plot no. 13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015

PROJECT PHOENIX FOR RE-NEET-2026

MM: 720 **CST - 02** **Time: 180 Mins.**

Answers

1. (1)	37. (3)	73. (1)	109. (3)	145. (1)
2. (3)	38. (2)	74. (2)	110. (2)	146. (2)
3. (3)	39. (1)	75. (4)	111. (3)	147. (1)
4. (3)	40. (4)	76. (2)	112. (4)	148. (1)
5. (4)	41. (4)	77. (2)	113. (1)	149. (1)
6. (1)	42. (3)	78. (1)	114. (1)	150. (4)
7. (1)	43. (2)	79. (1)	115. (1)	151. (3)
8. (1)	44. (2)	80. (1)	116. (3)	152. (4)
9. (2)	45. (2)	81. (2)	117. (1)	153. (3)
10. (1)	46. (2)	82. (4)	118. (4)	154. (4)
11. (3)	47. (2)	83. (3)	119. (2)	155. (3)
12. (1)	48. (1)	84. (2)	120. (4)	156. (4)
13. (3)	49. (2)	85. (2)	121. (2)	157. (3)
14. (2)	50. (3)	86. (2)	122. (1)	158. (3)
15. (2)	51. (4)	87. (4)	123. (3)	159. (1)
16. (1)	52. (4)	88. (2)	124. (1)	160. (3)
17. (1)	53. (3)	89. (2)	125. (1)	161. (2)
18. (2)	54. (2)	90. (2)	126. (4)	162. (1)
19. (2)	55. (2)	91. (1)	127. (3)	163. (2)
20. (4)	56. (2)	92. (4)	128. (1)	164. (3)
21. (4)	57. (4)	93. (4)	129. (1)	165. (4)
22. (3)	58. (4)	94. (1)	130. (3)	166. (4)
23. (4)	59. (2)	95. (2)	131. (3)	167. (1)
24. (4)	60. (4)	96. (2)	132. (4)	168. (2)
25. (3)	61. (4)	97. (3)	133. (3)	169. (1)
26. (1)	62. (2)	98. (2)	134. (4)	170. (1)
27. (2)	63. (1)	99. (4)	135. (4)	171. (2)
28. (4)	64. (4)	100. (2)	136. (3)	172. (4)
29. (1)	65. (3)	101. (3)	137. (1)	173. (2)
30. (4)	66. (3)	102. (3)	138. (3)	174. (1)
31. (3)	67. (1)	103. (3)	139. (3)	175. (4)
32. (3)	68. (4)	104. (2)	140. (2)	176. (3)
33. (2)	69. (4)	105. (2)	141. (4)	177. (1)
34. (4)	70. (4)	106. (3)	142. (3)	178. (3)
35. (4)	71. (1)	107. (4)	143. (4)	179. (4)
36. (2)	72. (3)	108. (2)	144. (4)	180. (1)

Hints and Solutions

PHYSICS

(1) **Answer :** (1)
Hint:
$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$; to find $\vec{v}_A$ **, use integration**
Solution:

$$\vec{v}_A = \int_0^t a_A dt = \int_0^2 2t dt = \left[t^2\right]_0^2 = 4\hat{i}$$

$$\vec{v}_B = \vec{a}_B t = 4 \times 2 = 8\hat{j}$$

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B = 4\hat{i} - 8\hat{j}$$

$$\left|\vec{v}_{AB}\right| = \sqrt{(4)^2 + (8)^2} = \sqrt{16+64} = \sqrt{80}$$

(2) **Answer :** (3)
Solution:
Using $v^2 - u^2 = 2as$

$$\frac{u^2}{4} - u^2 = 2a\left(30\right) \quad -\left(1\right)$$

$$0 - \frac{u^2}{4} = 2a\left(s\right) \quad -\left(2\right)$$

Dividing (1) by (2):

$$\frac{\frac{-3u^2}{4}}{\frac{-u^2}{4}} = \frac{30}{s}$$

$$\Rightarrow 3 = \frac{30}{s} \Rightarrow s = 10 \text{ cm}$$

$\therefore$ length of block is $30 + 10 = 40$ cm

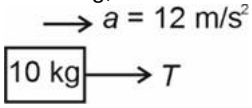
(3) **Answer :** (3)
Solution:
$u = 0$
$a = 0.5 \text{ m/s}$

$$S_{7^{\text{th}} \text{ half second}} = S_{3.5} - S_3$$

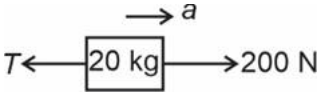
$$= \frac{1}{2}a\left[(3.5)^2 - (3)^2\right]$$

$$= \frac{1}{2} \times \frac{1}{2}\left[12.25 - 9\right] = 0.8125 \text{ m}$$

(4) **Answer :** (3)
Solution:
For 10 kg,



$T = 10 \times 12 = 120$ N for 20 kg,



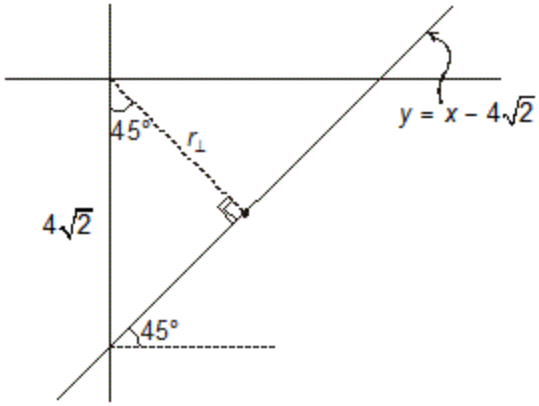
200 – T = 20a
200 – 120 = 20a
80 = 20a ⇒ a = 4 m/s^2

(5) **Answer :** (4)
Solution:
For a rigid body, separation between any two points does not change.
For rotation about a fixed axis, $\vec{\omega}$ has a fixed direction.

(6) **Answer :** (1)

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Solution:

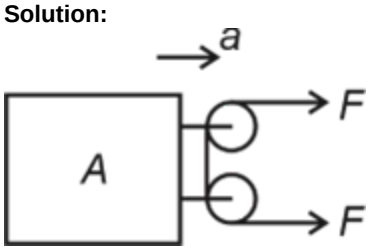


From figure, $r_\perp = 4\sqrt{2}\cos 45^\circ = 4$

Now, $L = mvr_\perp$ = (0.3)(20)(4) = 24 kg m^2s^{-1}

(7) **Answer :** (1)

Solution:



$a = \frac{20+20}{10} = 4$ m/s^2

(8) **Answer :** (1)

Solution:

mgh = μ mgx

$x = \frac{h}{\mu} = \frac{1}{0.2} = 5\ m$

Hence, after to and fro motion once it will cover 4 m. Therefore, block will be 1 m away from *A*.

(9) **Answer :** (2)

Hint:

$\Delta U = U_f - U_i$

Solution:

$U_i = -\frac{GMm}{R+h}$

$U_i = -\frac{GMm}{R+nR}$; $U_f = \frac{-GMm}{R}$.

Change in gravitational potential energy is

$\Delta U = U_{\text{surface}} - U_{\text{height}}$

$= -\frac{GMm}{R} - \left[-\frac{GMm}{R(n+1)}\right]$

$= \frac{GMm}{R}\left[\frac{1}{n+1} - 1\right] = -\frac{GMm}{R}\left(\frac{n}{n+1}\right)$

$= \frac{-n}{n+1}(mgR).$

(10) **Answer :** (1)

Hint:

int: Terminal velocity, $V_T = \frac{2}{9}\frac{r^2(\rho-\sigma)}{\eta}g$

Solution:

$\eta = \frac{2}{9} \times \frac{(10^{-3})^2\,\text{m}^2[4.2\times10^3 - 1.2\times10^3]\ \text{kg/ m}^3}{2\times10^{-2}\ \text{ms}^{-1}} \times 10\ \text{m/s}^2$

$\eta = 3.3 \times 10^{-1}$ kg $m^{-1}s^{-1}$

(11) **Answer :** (3)

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Hint:
For adiabatic process *PV*γ = constant
Solution:
PV = *nRT*
$V = \frac{nRT}{P}$
*PV*γ = constant
$P\left(\frac{nRT}{P}\right)^{\gamma}$ = constant
$P \propto (T)^{\frac{\gamma}{\gamma-1}}$
$\Rightarrow \frac{\gamma}{\gamma-1} = k$
and $\gamma = \frac{7}{5}$
$k = \frac{7}{2}$

(12) Answer : (1)
Solution:
$R_{\text{net}} = \frac{R_1 R_2}{R_1+R_2}$
$R_1 = \frac{l}{k\pi R^2}$, $R_2 = \frac{l}{2k8\pi R^2}$
$R_{\text{net}} = \frac{\frac{l}{k\pi R^2} \times \frac{l}{16\pi kR^2}}{\frac{l}{k\pi R^2} + \frac{l}{16\pi R^2 K}}$
$\frac{l}{9\pi R^2 K_{\text{eff}}} = \frac{\frac{l}{k\pi R^2} \times \frac{l}{16\pi kR^2}}{\frac{l}{k\pi R^2} + \frac{l}{16\pi R^2 k}}$
$\frac{1}{9K_{\text{eff}}} = \frac{\frac{1}{16K}}{1+\frac{1}{16}}$
$9K_{\text{eff}} = 17K$
$K_{\text{eff}} = \frac{17K}{9}$

(13) Answer : (3)
Solution:
Body moves under constant acceleration may moves on straight line or on curve path.

(14) Answer : (2)
Hint:
$h = \frac{2S\cos\theta}{\rho g r}$
Solution:
$h' = \frac{2S\cos\theta}{\rho g_{\text{eff}} r}$
$h' = \frac{2S\cos\theta}{\rho \frac{g}{2} r}$
$h' = \frac{4S\cos\theta}{\rho g r} = 2 \times 8 = 16$ cm

(15) Answer : (2)
Solution:
Thermal strain = Longitudinal strain = α Δ *T*
⇒ Longitudinal strain, δ = $10^{-5} \times 10^{2} = 10^{-3}$
⇒ Compressive stress = δ × Young's Modulus
$= 10^{-3} \times 0.5 \times 10^{11}$
$= 0.5 \times 10^{8}$
⇒ Compressive force = $0.5 \times 10^{8} \times 10^{-3} = 0.5 \times 10^{5}$
$= 5 \times 10^{4} \times \frac{10}{10}$
$= 50 \times 10^{3}$ N

(16) Answer : (1)
Solution:
$W = 2P_0 \times 2V_0 = 4P_0V_0$

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In process *AB*, heat supplied = $nC_v\Delta T = n \times \frac{3}{2}R \times 2T_0$

= 3 nRT_0

In process *BC*, heat supplied = $nC_p\Delta T = n \times \frac{5}{2}R6T_0$

= 15 nRT_0

Total heat supplied = $18nRT_0 = 18P_0V_0$

$\eta = \frac{W}{Q_{\text{sup plied}}} \times 100 = \frac{4P_0V_0}{18P_0V_0} \times 100 = 22.2\%$

(17) **Answer :** (1)

Solution:

Heat gained = $ms_{\text{ice}}\Delta\theta + mL + ms_{\text{water}}\Delta\theta$

$= 5 \times 0.5 \times [0-(-10)] + 5 \times 80 + 5 \times 1 \times (\theta - 0)$

$= 25 + 400 + 5\theta$

Heat loss $= ms_{\text{water}}\Delta\theta$

$= 20 \times 1 \times (60 - \theta)$

Heat gain = Heat loss

$425 + 5\theta = 1200 - 20\theta \Rightarrow 25\theta = 775$

$\theta = 31^\circ\text{C}$

(18) **Answer :** (2)

Hint:

$f_A - f_B = 5$

Solution:

$\left(f_C + \frac{5f_c}{100}\right) - \left(f_C - \frac{3f_c}{100}\right) = 5$

$5f_c + 3f_c = 500$

$8f_c = 500$

$f_c = 62.5$ Hz

$f_B = 62.5 - 3 \times \frac{62.5}{100}$

= 60.625 Hz

(19) **Answer :** (2)

Hint:

$P_{\text{avg}} = \frac{1}{2}\mu v\omega^2 A^2$

Solution:

$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{90}{0.1}} = 30\,m/s$

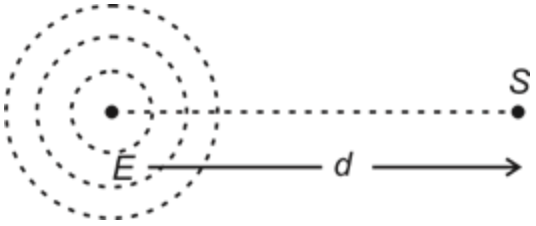
$P_{\text{avg}} = \frac{1}{2} \times 0.1 \times 30 \times (100)^2 \times 10^{-4} = \frac{3}{2}$W

(20) **Answer :** (4)

Hint:

Both waves travels with different speed, hence they reach at different instants

Solution:



$d = v_1t_1 = v_2t_2$

$6 \times t_1 = 12 \times t_2$

$t_1 = 2t_2$

$t_1 - t_2 = 3$ min $= 3 \times 60 = 180$ s

$2t_2 - t_2 = 180$ s

$\therefore t_2 = 180$ s

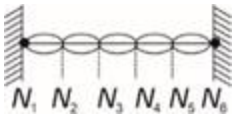
$\therefore d = v_2t_2 = 12 \times 180 = 2160$ km

(21) **Answer :** (4)

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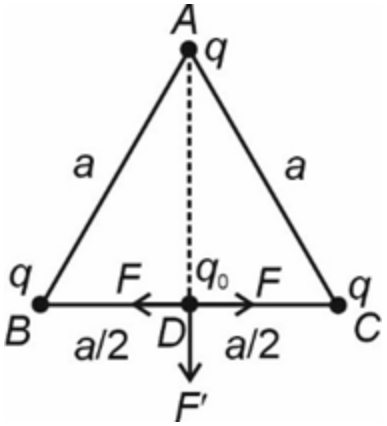
Hint:
Both masses will apply equal and opposite forces on each other.
Solution:
$\overrightarrow{F}_{12} = -\overrightarrow{F}_{21}$
$\Rightarrow \overrightarrow{F}_{12} + \overrightarrow{F}_{21} = 0$
$\Rightarrow \left|\overrightarrow{F}_{12}\right| = \left|\overrightarrow{F}_{21}\right|$

(22) Answer : (3)
Solution:
$\frac{n_1+n_2}{\gamma_{mix}-1} = \frac{n_1}{\gamma_1-1} + \frac{n_2}{\gamma_2-1}$
$\frac{6}{\gamma_{mix}-1} = \frac{2}{\frac{7}{5}-1} + \frac{4}{\frac{5}{3}-1}$
$\frac{6}{\gamma_{mix}-1} = \frac{2\times5}{2} + \frac{4\times3}{2}$
$\frac{6}{\gamma_{mix}-1} = 5+6 = 11 \Rightarrow \eta_{mix} = \frac{6}{11}+1 = \frac{17}{11} = 1.54$

(23) Answer : (4)
Solution:


Position of nodes at x (in m)
= 0, 0.05, 0.1, 0.15, 0.2, 0.25
Position of anti-nodes at x (in m)
= 0.025, 0.075, 0.125, 0.175, 0.225
$y = 0.01 \sin(62.8\,x)\cos(628\,t)$
On comparing it with $y = 2A \sin Kx \cos \omega t$
$2A = 0.01\text{ m},\ K = 62.8\text{ m}^{-1}$ and $\omega = 628\text{ s}^{-1}$
$\lambda = \frac{2\pi}{K} = \frac{2\pi}{62.8} = 0.1\text{ m}$
$L = 5\frac{\lambda}{2} = \frac{5}{2}(0.1) = 0.25\text{ m}$

(24) Answer : (4)
Solution:
For series combination $R_{Net} = R + R = 2R$
$P_1 = \frac{V^2}{2R}$
For parallel
$R_{Net} = \frac{R}{2}$
$P_2 = \frac{V^2}{\left(\frac{R}{2}\right)} = \frac{2V^2}{R} = 4P_1 = 400\text{ W}$.

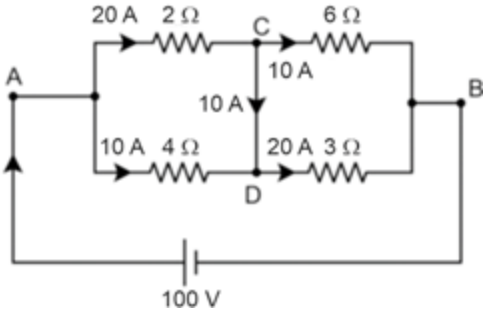
(25) Answer : (3)
Solution:


From geometry

$AD = \frac{\sqrt{3}}{2}a$

$F_{net} = F' = \frac{Kqq_0}{\left(\frac{\sqrt{3}}{2}a\right)^2} = \frac{4Kqq_0}{3a^2}$

(26) Answer : (1)
Solution:



$R_{AB} =$ (2 Ω and 4 Ω) in series with (6 Ω and 2 Ω)

$= \frac{2 \times 4}{2+4} + \frac{6 \times 3}{6+3} = \frac{4}{3} + 2 = \frac{10}{3}\Omega$

$i = \frac{100}{R_{AB}} = \frac{100 \times 3}{10} = 30\text{A}$

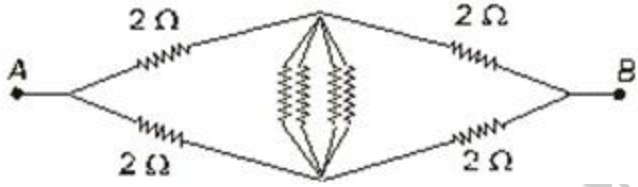
$i_{2\Omega} = \frac{4}{6} \times i = \frac{2}{3} \times 30 = 20$ A

$i_{4\Omega} = \frac{2}{6}i = \frac{1}{3} \times 30 = 10$ A

$i_{6\Omega} = \frac{3}{9} \times i = \frac{1}{3} \times 30 = 10$ A

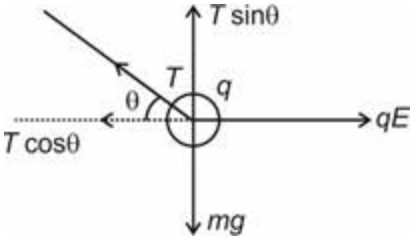
$i_{3\Omega} = 20$ A

(27) Answer : (2)
Hint:
Balanced Wheatstone bridge.
Solution:
Equivalent circuit is



$\therefore R_{AB} = 2\ \Omega$

(28) Answer : (4)
Solution:



At equilibrium
$T \sin\theta = mg$
$T \cos\theta = qE$

$\tan\theta = \frac{mg}{qE} = \frac{2 \times 10^{-3} \times 10}{8 \times 10^{-6} \times 10^3}$

$\tan\theta = \frac{5}{2}$

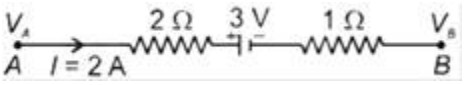
(29) Answer : (1)
Solution:

$F = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{9 \times 10^{-4}}$

$F = 10^{13} \times 256 \times 10^{-40}$

$F = 2.56 \times 10^{-25}$ N

(30) **Answer :** (4)
Solution:



$V_A - V_B = (2 \times 2) + 3 + (2 \times 1)$

$= 4 + 3 + 2$

$= 9\text{V}$

(31) **Answer :** (3)
Solution:

$C_{AB} = \frac{3\varepsilon_0 A}{d}$ $C_{XY} = \frac{\varepsilon_0 A}{3d}$

$\Rightarrow \quad \frac{C_{A'B}}{C_{XY}} = \frac{9}{1}$

(32) **Answer :** (3)
Solution:

Output $Y = \overline{(A.B).C} = \overline{A} + \overline{B} + \overline{C}$

When A,B,C are $0 \rightarrow Y = 1$

When A,B,C are $1 \rightarrow Y = 0$

(33) **Answer :** (2)
Solution:

$\frac{E_{rms}}{B_{rms}} = c$

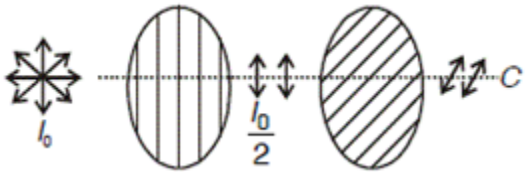
$B_{rms} = \frac{E_{rms}}{c} = \frac{6}{3 \times 10^8}$

$B_{rms} = 2 \times 10^{-8}$

$B_{rms} = \frac{B_0}{\sqrt{2}}$

$B_0 = \sqrt{2} \times B_{rms} = \sqrt{2} \times 2 \times 10^{-8} = 2.83 \times 10^{-8} T$

(34) **Answer :** (4)
Solution:



For 1st polaroid

$I_1 = I_0 \cos^2\theta$

Take average value of $\cos^2\theta = \frac{1}{2}$

$I_1 = \frac{I_0}{2}$

For 2nd polaroid

$I_2 = I_1 \cos^2\theta'$

$= \frac{I_0}{2} \times \cos^2 30^\circ = \frac{3I_0}{8}$

(35) **Answer :** (4)
Solution:

$n_1\lambda_1 = n_2\lambda_2$

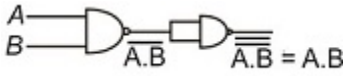
(36) **Answer :** (2)
Solution:

$\lambda = \frac{h}{\sqrt{3mKT}}$

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As temperature doubled, so $\lambda' = \frac{\lambda}{\sqrt{2}}$

(37) Answer : (3)
Solution:
Apparent depth from water surface = $\frac{x}{\mu}$.
From observer = $h + \frac{x}{\mu}$

(38) Answer : (2)
Solution:


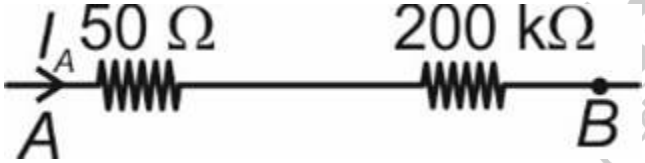
2 AND gate required.

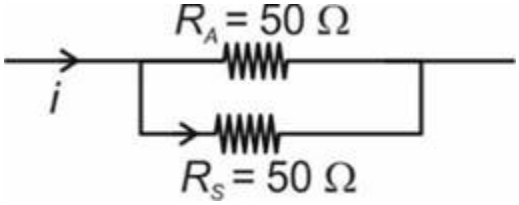
(39) Answer : (1)
Solution:
$\frac{1}{f_1} = (\mu_g - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$ ...(1)
$\frac{1}{f_2} = \left(\frac{\mu_g}{\mu_w} - 1\right)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$ ...(2)

$\frac{\text{Eq}^{n}\,(1)}{\text{Eq}^{n}\,(2)}\ \frac{f_2}{f_1} = \frac{\mu_g - 1}{\left(\frac{\mu_g}{\mu_w} - 1\right)}$

$\frac{f_2}{f_1} = \frac{\left(\frac{3}{2} - 1\right)}{\left(\frac{\frac{3}{2}}{\frac{4}{3}} - 1\right)}$

$f_2 = 4\, f_1$
$f_2 = 80$ cm

(40) Answer : (4)
Solution:
If R_S = 200 k Ω in series
Then


$V_A - V_B \simeq I_A \times 200$ k Ω
= 50 × 10^{-6} × 200 × 10^3
$\simeq$ 10 V
So range of volt meter = 10 V
If R_S = 50 Ω in parallel
Then


As R_A and R_S in parallel and $R_A = R_S$ so current in both have same value
So $i = 2\, I_A$ = 50 × 10^{-6} × 2
= 100 × 10^{-6} A
= 0.1 mA

(41) Answer : (4)
Solution:
$\lambda = \frac{h}{\sqrt{2mqV}}$, $\frac{\lambda_P}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_P q_P}} = \sqrt{\frac{8}{1}} = 2\sqrt{2} : 1$

(42) Answer : (3)
Solution:
Use $\beta = \frac{\lambda D}{d}$
$\beta' = \frac{\lambda \times 3D}{\frac{d}{2}} = 6\beta$

(43) Answer : (2)
Solution:
According to Einstein's photoelectric equation
$K_{\max} = eV_0 = h\nu - \phi_0$
$V_0 = \frac{h}{e}\nu - \frac{\phi_0}{e}$
$\therefore$ *h* = slope of line × *e*

(44) Answer : (2)
Solution:
$\Delta E = \frac{hc}{\lambda}$
$\lambda = \frac{hc}{\Delta E} = \frac{hc}{\frac{3}{2}E} = \frac{2hc}{3E}$

(45) Answer : (2)
Solution:
L.C = MSD – VSD ...(1)
given : (*N* + 1) VSD = *N* MSD
$\text{VSD} = \left(\frac{N}{N+1}\right)\ \text{MSD}$ ...(2)
From (1) and (2)
$L.C = (\text{MSD}) - \frac{N}{N+1}\left(\text{MSD}\right)$
$= \text{MSD}\ \left(1 - \frac{N}{N+1}\right) = \frac{\text{MSD}}{N+1}$
$= \frac{0.01}{N+1} = \frac{1}{100(N+1)}$

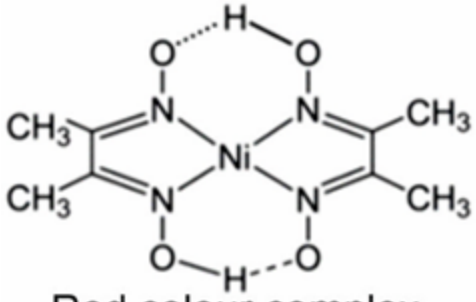
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CHEMISTRY

(46) Answer : (2)
Solution:
• Zr(160 pm) and Hf (159 pm) have almost identical radii
• $Ce^{4+}(f^0)$ and Yb^{2+} (f^{14}) have no unpaired electrons so they are diamagnetic in nature.

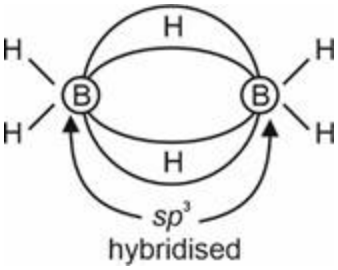
(47) Answer : (2)
Hint:
Higher the solubility, lower will be the precipitation
Solution:
K_{sp} [AgI] = $[Ag^+]$ $[I^-]$
$[I^-] = 8.5 \times 10^{-16}$ M
K_{sp} $[Hg_2I_2] = [Hg_2^{2+}][I^-]^2$
$[I^-] = 5 \times 10^{-13}$ M
$[I^-]$ for AgI is smaller than for Hg_2I_2
$\therefore$ At 5×10^{-13} M concentration most of the AgI has gone in precipitate.

(48) Answer : (1)
Solution:
Nickel sulphide dissolves in aqua regia
$3\text{NiS} + 2\,\text{HNO}_3 + 6\text{HCl} \rightarrow 3\,\text{NiCl}_2 + 2\text{NO} + 3\text{S} + 4\text{H}_2\text{O}$
$2\,\text{CuSO}_4 + \text{K}_4\left[\text{Fe}\left(\text{CN}\right)_6\right] \rightarrow \underset{\text{Chocolate brown ppt}}{\text{Cu}_2[\text{Fe}(\text{CN})_6]} + 2\text{K}_2\,\text{SO}_4$



(49) **Answer :** (2)
Solution:

Each boron atoms in diborane uses sp^3 hybrid orbitals for bonding.



(50) **Answer :** (3)
Solution:
Molar conductivity increases on dilution of electrolyte solution.

(51) **Answer :** (4)
Solution:

(Vitamins)	(Deficiency Diseases)
Thiamine	– Beri-beri
Riboflavin	– Cheilosis
Pyridoxine	– Convulsion
Ascorbic acid	– Scurvy

(52) **Answer :** (4)
Solution:

$E_{H^+/H_2} = E^o_{H^+/H_2}$

$- \frac{0.0591}{2} \log \frac{P_{H_2}}{[H^+]^2}$

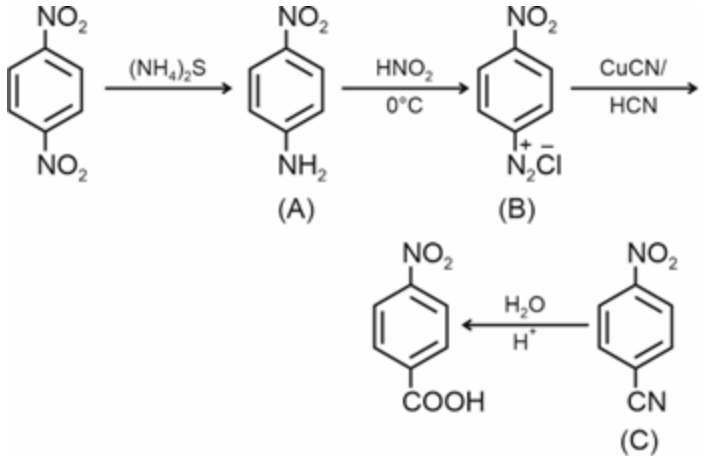
$\therefore E_{H^+/H_2} = 0 - \frac{0.0591}{2} \log \frac{1}{(10^{-2})^2}$

$= - \frac{0.0591}{2} \times 4 = -0.118 \text{ V}$

(53) **Answer :** (3)
Solution:
Monosaccharides cannot be hydrolysed to give poly-hydroxy aldehydes and ketones.
The two six-membered cyclic hemiacetal forms of D–(+)– glucose are called anomers.
Hydrolysis of sucrose gives dextrorotatory glucose and laevorotatory fructose.

(54) **Answer :** (2)
Solution:

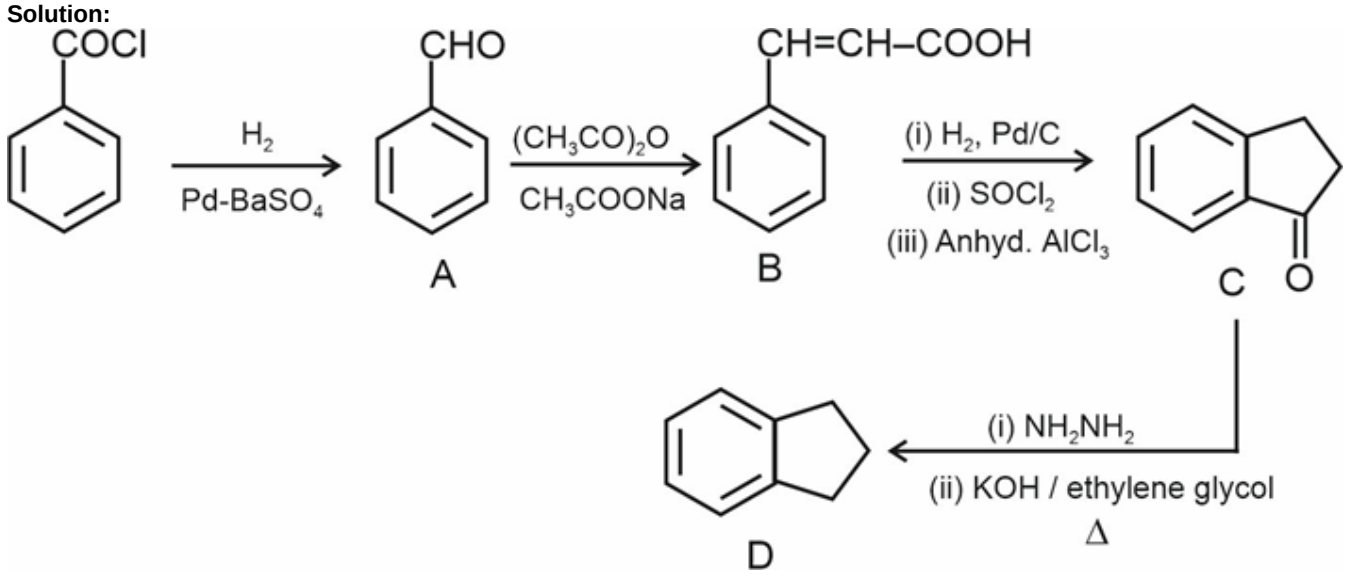
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(55) Answer : (2)
Solution:

$$2\,CH_3\,COOH \xrightarrow[573\,K]{MnO} CO_2 + H_2O + CH_3\,COCH_3$$

(56) Answer : (2)
Solution:



(57) Answer : (4)
Hint:

Percentage of nitrogen = $\frac{28 \times \text{volume of } N_2 \text{ at STP} \times 100}{22400 \times \text{mass of organic compound}}$

Solution:
V_1 = 30 mL
P_1 = 715 – 15 = 700 mm Hg
T_1 = 300 K

So, $V_2 = \frac{P_1 \times V_1 \times 273}{T_1 \times 760} = \frac{700 \times 30 \times 273}{300 \times 760}$

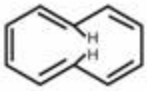
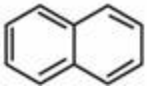
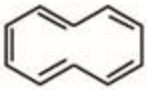
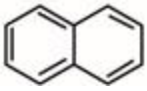
= 25.1 mL

Percentage of nitrogen = $\frac{28 \times 25.1 \times 100}{22400 \times 0.1}$

= 31.4%

(58) Answer : (4)
Hint:
Cyclic, planar organic compounds containing (4 n + 2) π electrons in conjugated system are aromatic in nature.

Solution:

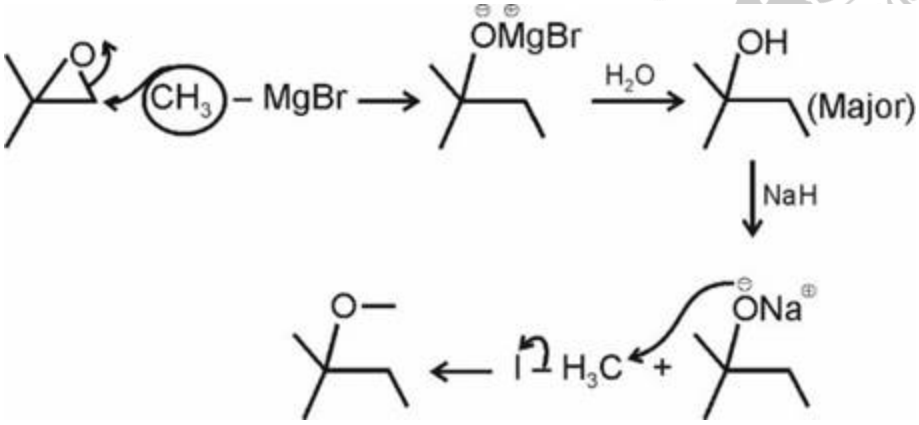
-  : Non-aromatic (tube shaped)
-  : Non-aromatic because of steric hindrance of the H-atoms shown
-  : Antiaromatic
-  : Aromatic
-  : Non-aromatic (tube shaped)
-  : Non-aromatic
-  : Antiaromatic
-  : Aromatic

(59) **Answer :** (2)

Hint:

Grignard reagent preferably attacks by S_N2 mechanism at sterically less hindered carbon of the epoxide.

Solution:



(60) **Answer :** (4)

Hint:

Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion.

Solution:

Complex	Magnetic character	Isomerism
$[Cr(NH_3)_4Br_2]Br$	Cr^{3+} is d^3, hence paramagnetic	Cis-trans
$[Ti(H_2O)_5Br](NO_3)_2$	Ti^{3+} is d^1, hence paramagnetic	Ionization
$[Pt(en)(NH_3)Br]NO_3$	Pt^{2+} is d^8, complex is square planar, all electrons are paired	Ionization
$[Co(NH_3)_4(NO_3)_2]NO_3$	Co^{3+} is d^6, hence diamagnetic	Cis-trans

(61) **Answer :** (4)

Solution:

$2KMnO_4 + 10\ KI + 8H_2SO_4 \rightarrow 6K_2SO_4 + 2MnSO_4 + 5I_2 + 8H_2O$

So 10 moles of KI require = 2 moles of $KMnO_4$

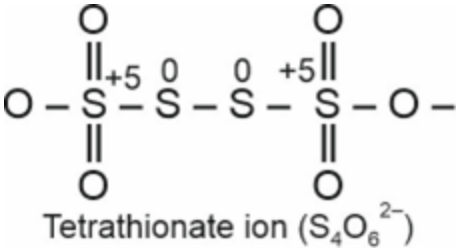
1 mole of KI require $= \frac{2}{10} = \frac{1}{5}$

(62) **Answer :** (2)
Solution:
Ba^{2+} is group V cation while Ni^{2+} is group IV cation.
Group reagent for group V – $(NH_4)_2CO_3$ in presence of NH_4OH.
Group reagent for group IV – H_2S in presence of NH_4OH.
Brown ring test is the confirmatory test of NO_3^-ion.

(63) **Answer :** (1)
Hint:
Species having unpaired electrons are paramagnetic in nature.
Solution:
Electronic configuration of O_2 according to MOT
$O_2 : \sigma 1s^2,\ \sigma * 1s^2,\ \sigma 2s^2,\ \sigma * 2s^2,\ \sigma 2p_z^2,$
$\pi 2p_x^2 = \pi 2p_y^2,\ \pi * 2p_x^1 = \pi * 2p_y^1$
Since O_2 has two unpaired electron in π* molecular orbitals it will be paramagnetic in nature.

(64) **Answer :** (4)
Solution:
$E_{(2s)}H > E_{(2s)}Li > E_{(2s)}Na$
Energies of the orbitals in the same subshell decrease with increase in the atomic number
Angular node = ℓ
For 3p orbital angular nodes = 1

(65) **Answer :** (3)
Solution:



Tetrathionate ion ($S_4O_6^{2-}$)

(66) **Answer :** (3)
Solution:
During adiabatic expansion of an ideal gas, cooling takes place.
$\therefore T_C < T_A$
And $w_{isothermal} > w_{adiabatic}$
During isothermal process $\Delta T = 0$
$\therefore T_A = T_B$

(67) **Answer :** (1)
Solution:
$XeF_6 + NaF \rightarrow Na^+[XeF_7]^-$

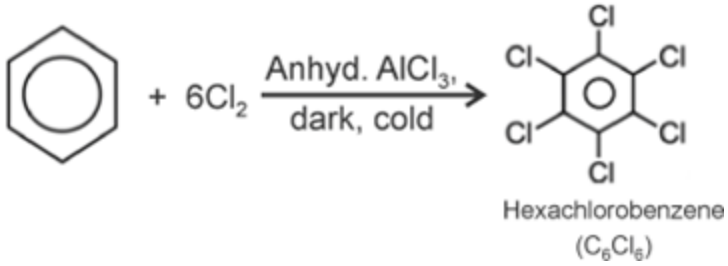
(68) **Answer :** (4)
Solution:
Availability of lone pair is large.

(69) **Answer :** (4)
Solution:
In group 16, the least negative value of electron gain enthalpy is associated with Oxygen.

Element	$\Delta_{eg}H\ (kJ\ mol^{-1})$
F	–328
S	–200
Se	–195
O	–141

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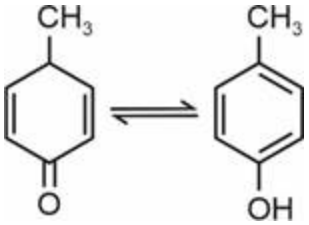
(70) Answer : (4)
Solution:

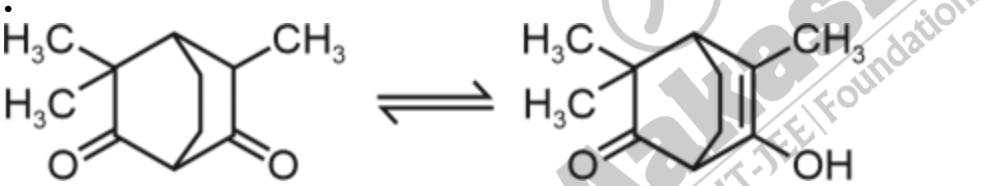


Hexachlorobenzene
(C_6Cl_6)

(71) Answer : (1)
Solution:
The oxidation number of Cl in ClO_2^- is +3 which is the intermediate oxidation state for Cl and it can be oxidised and reduced simultaneously.

(72) Answer : (3)
Hint:
Bridgehead carbon cannot be sp^2 hybridized
Solution:





(73) Answer : (1)
Solution:
A catalyst catalyses the spontaneous reactions but does not catalyse non-spontaneous reactions.

(74) Answer : (2)
Solution:

$\Delta S = 2.303 \ \ nR \ \log \frac{V_2}{V_1}$

$= 2.303 \times 1 \times 2 \log \frac{20}{2}$

$= 2.303 \times 2 \times \log 10 = 4.606 \ calK^{-1}mol^{-1}$

(75) Answer : (4)
Solution:
For concentration cell, $E^o_{cell} = 0$

$E_{cell} = -\frac{2.303RT}{nF} \log \frac{C_1}{C_2}$

For a spontaneous cell reaction $C_2 > C_1$

For cell reaction:

Zn(s) | Zn^{2+}(0.01 M) || Zn^{2+}(0.1 M) | Zn(s)

$C_2 > C_1$

$\therefore \ E_{cell} = -\frac{2.303}{2F} \log \frac{0.01}{0.1}$

$= -\frac{0.0591}{2} \log 10^{-1}$

E_{cell} = +0.0295

(76) Answer : (2)

Solution:

A	+	2B	→	3C
2		2		0
1		0		3

3 moles of c is formed.
1 mole of A is left after the reaction.

(77) **Answer :** (2)
Hint:

Electronic configuration	Element
$1s^2 2s^2$	Be
$1s^2 2s^2 2p^1$	B
$1s^2 2s^2 2p^3$	N
$1s^2 2s^2 2p^4$	O

Solution:
Due to stable electronic configuration of 'Be' & 'N', they have higher first ionization enthalpy than B and O respectively

B < Be < C < O < N < F

(801 889 1086 1314 1402 1681) $\Delta_i H$ / kJ mol^{-1}

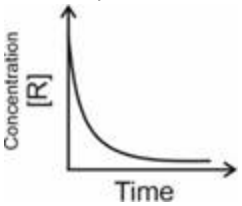
(78) **Answer :** (1)
Solution:
Oxidation state of metal tells about primary valency while the coordination number tells about secondary valency.

(79) **Answer :** (1)
Solution:
Diffraction and interference are explained by wave nature of electromagnetic radiations.

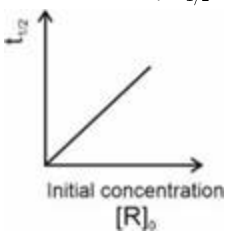
(80) **Answer :** (1)
Hint:
Removal of SCN^-(aq) ion shifts the equilibrium from right to left.
Solution:
Hg^{2+} reacts with SCN^-(aq) ions to form a stable complex $[Hg(SCN)_4]^{2-}$(aq).
Removal of free SCN^-(aq) ions shifts equilibrium backward and intensity of red colour due to $[Fe(SCN)]^{2+}$ decreases.

(81) **Answer :** (2)
Solution:
Higher the value of K_H, lower is the solubility.
So, correct order of solubility is He < N_2 < H_2 < O_2

(82) **Answer :** (4)
Solution:
$[R] = [R]_0 e^{-kt}$



Follow first order kinetics
For zero order; $t_{1/2} = [R]_0/2k$



(83) **Answer :** (3)

Solution:

Co^{3+} (high spin)

d^6 configuration = $t_{2g}^4 e_g^2$ (unsymmetrical)

Fe^{3+} (low spin)

d^5 configuration = $t_{2g}^5 e_g^0$ (unsymmetrical)

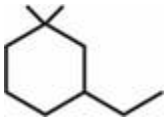
so these complexes show Jahn Teller distortion.

(84) **Answer :** (2)

Hint:

A saturated monocyclic compound is named by prefixing 'cyclo' to the corresponding straight chain alkane.

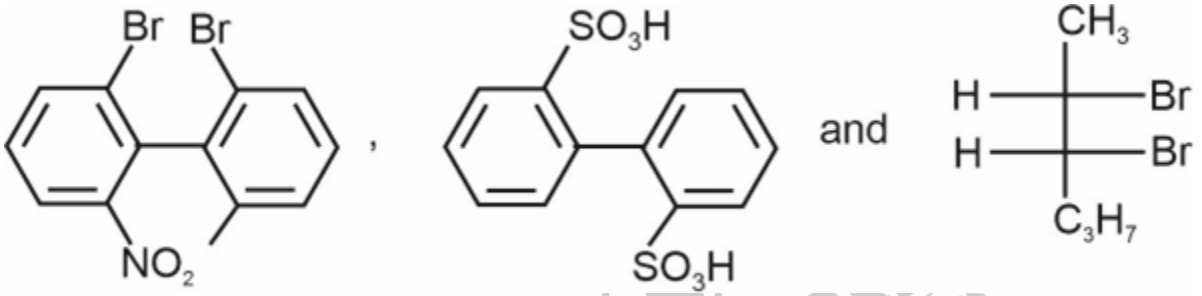
Solution:



3-Ethyl-1,1-dimethylcyclohexane

(85) **Answer :** (2)

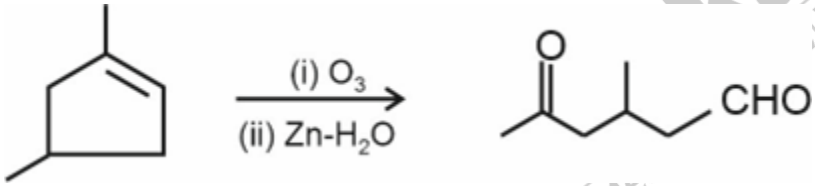
Solution:



do not contain plane of symmetry so these are optically active compounds.

(86) **Answer :** (2)

Solution:



(87) **Answer :** (4)

Solution:

Since $E_{a_2} > E_{a_1}$

$\therefore$ B → C is the rate determining step.

(88) **Answer :** (2)

Solution:

$O_3 \rightleftharpoons O_2 + O$ (fast) ...(1)

$O + O_3 \rightarrow 2O_2$ (slow) ...(2)

From eq. (1), $K_{eq} = \frac{[O_2][O]}{[O_3]} \Rightarrow [O] = \frac{k_{eq}[O_3]}{[O_2]}$

From eq. (2), $r = k[O][O_3] = k\frac{k_{eq}[O_3][O_3]}{[O_2]}$

$r = k'[O_3]^2[O_2]^{-1}$

Since it is a first order reaction overall hence unit of rate constant = sec^{-1}

(89) **Answer :** (2)

Solution:

Metaphosphophoric acid $(HPO_3)_n$	– $H_3PO_3 + Br_2$, heated in a sealed tube.
Phosphonic acid	– $P_2O_3 + H_2O$
Hypophosphoric acid	– Red P_4 + alkali

Pyrophosphorous acid – $PCl_3 + H_3PO_3$

(90) Answer : (2)
Hint:

$$A(s) \rightarrow \underset{p_1}{B(g)} + \underset{p_1+p_2}{C(g)}$$

$$D(s) \rightarrow \underset{p_1+p_2}{C(g)} + \underset{p_2}{E(g)}$$

Solution:
$K_{p_1} = (p_1)(p_1 + p_2)$
$a = p_1^2 + p_1p_2$ ...(i)
$K_{p_2} = (p_2)(p_1 + p_2)$
$b = p_2^2 + p_1p_2$ ...(ii)
$a + b = p_1^2 + 2p_1p_2 + p_2^2$
$(a + b) = (p_1 + p_2)^2$ ...(iii)
Calculating total pressure:
$p_T = p_B + p_C + p_E$
$= p_1 + p_1 + p_2 + p_2$
$= 2p_1 + 2p_2$
$= 2(p_1 + p_2)$
Putting the value of $p_1 + p_2$ in equation (iii)
$p_T = 2\left(\sqrt{a + b}\right) atm$

BIOLOGY

(91) Answer : (1)
Solution:
Pyrenoids are found in the chloroplasts of most of the green algae.
Pyrenoids are storage bodies in chloroplasts of most of the members of Chlorophyceae. These contain protein and starch.

(92) Answer : (4)
Solution:
'X' represents the epicotyl hook.

(93) Answer : (4)
Solution:
Chalaza represents the basal part of the ovule

(94) Answer : (1)
Solution:
Musca domestica – Diptera
Mangifera indica – Sapindales
Solanum melongena – Polymoniales
Triticum aestivum – Poales

(95) Answer : (2)
Solution:
Gibberellins are the terpene derivatives, not the indole compounds.

(96) Answer : (2)
Solution:

- The NADH + H^+ is reoxidised to NAD^+ in both types of fermentation.
- CO_2 and ethanol are produced in alcoholic fermentation.
- Not much energy is released in fermentation.
- Pyruvate decarboxylase is used in alcoholic fermentation.

(97) Answer : (3)
Solution:
Statement A: All living organisms retain the enzymatic machinery for glycolysis. This statement is correct because glycolysis is a fundamental metabolic pathway present in almost all living organisms, providing energy through the breakdown of

glucose.
Statement B: In naturally fermented alcoholic beverages, maximum concentration of alcohol could be less than 13%. This statement is also correct. Naturally fermented beverages typically have an alcohol concentration that does not exceed 13% due to the tolerance limit of yeast to alcohol.
Therefore, the correct option is (3) Both A and B are correct.

(98) Answer : (2)
Solution:
The anatomy of seed plants by Katherine Esau was published in 1960.

(99) Answer : (4)
Solution:
Cuscuta on hedge plants is an example of ectoparasite.

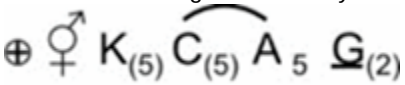
(100) Answer : (2)
Solution:
Species-Area relationship was first given by German naturalist and geographer Alexander von Humboldt.

(101) Answer : (3)
Solution:
Perispermic seeds are observed in black pepper, beet and castor.

(102) Answer : (3)
Solution:
Wine and beer are produced without distillation.

(103) Answer : (3)
Solution:
Out of four DNA strands, one would have ^{15}N.

(104) Answer : (2)
Solution:
Belladonna belongs to the family Solanaceae and the floral formula of this family is:



(105) Answer : (2)
Solution:

Bulliform cells	– Large, empty and colourless cells found in monocots
Cuticle	– Waxy thick layer prevents water loss
Stomata	– Composed of dumb-bell shaped guard cells and tiny pore in grasses
Epidermis	– Usually single layered and present in most plant parts

(106) Answer : (3)
Solution:
The epithelial tissue has a free surface which faces either a body fluid or outside environment. The cells are compactly packed with little intercellular matrix.

(107) Answer : (4)
Solution:
Epithelial tissues are avascular.
Neuroglial cells are not excitable cells.
Multicellular organisms exhibit division of labour.

(108) Answer : (2)
Solution:
Lipids (P) are not polymers. Nucleic acids (Y) are polymers of nucleotides and proteins (X) are polymers of amino acids. Enzymes (Z) are generally proteins.

(109) Answer : (3)
Solution:
There is one feature common to all those compounds found in the acid-soluble pool, *i.e.*, they have molecular weights ranging from 18 to around 800 dalton (Da) approximately.
The acid-insoluble fraction has only four types of organic compounds *i.e.*, proteins, nucleic acids, polysaccharides and lipids.
Chemical compounds found in living organisms which have molecular weight less than 1000 daltons are found in acid-soluble fraction and are called biomicromolecules.
Lipids are not strictly macromolecules.

(110) Answer : (2)

Solution:

The most abundant protein present in the animal world is collagen and it is arranged in an irregular pattern in dense irregular connective tissue, that is present beneath the skin.

(111) Answer : (3)

Hint:

Prosthetic group

Solution:

There are three kinds of co-factors–prosthetic groups, co-enzymes and metal ions.

Prosthetic groups are organic compounds and are distinguished from other co-factors in that they are tightly bound to the apoenzyme. For example, in peroxidase and catalase, which catalyse the breakdown of hydrogen peroxide to water and oxygen, haem is the prosthetic group and it is a part of active site of the enzyme. Zinc is a co-factor for the proteolytic enzyme carboxypeptidase.

(112) Answer : (4)

Solution:

In the presence of a competitive inhibitor, the K_m value increases and the V_{max} remains constant.

(113) Answer : (1)

Solution:

The pons region of the brain contains pneumotaxic centre that can moderate the functions of the respiratory rhythm centre. Neural signals from pneumotaxic centre can reduce the duration of inspiration and thereby alter the respiratory rate.

(114) Answer : (1)

Solution:

Vital capacity can be represented as: ERV + TV + IRV or TLC – RV

ERV + RV = Functional Residual Capacity

(115) Answer : (1)

Solution:

To obtain a standard ECG, a patient is connected to the machine with three electrical leads (one to each wrist and to the left ankle) that continuously monitor the heart activity.

For detailed evaluation of the heart's function, multiple leads are attached to the chest region.

(116) Answer : (3)

Solution:

$$\left.\begin{array}{r}\text{Round and green seeded pea} = 3\\ \text{Wrinkled and yellow seeded pea} = 3\end{array}\right] \; F_2 \text{ generation}$$

New combinations = 6/16 × 100 = 37.5%

(117) Answer : (1)

Solution:

LHC are made up of hundreds of pigment molecules bound to proteins.

(118) Answer : (4)

Hint:

Clarias gariepinus was introduced in India for aquaculture purposes.

Solution:

Lantana is a serious terrestrial weed and has replaced many species in forest.

Nile perch is responsible for elimination of over 200 species of cichlid fish in Lake Victoria of East Africa.

(119) Answer : (2)

Solution:

Ladybird and Dragonflies are useful to get rid of aphids and mosquitoes.

(120) Answer : (4)

Solution:

Drones (males) do not have father and thus cannot have sons, but have a grandfather and can have grandsons.

(121) Answer : (2)

Hint:

In sub-metacentric chromosome, centromere is present slightly away from the centre.

Solution:

In metacentric chromosome, centromere is present at the centre and is also called isobrachial chromosome.

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In acrocentric chromosome, centromere is present very close to one end of chromosome. Telocentric chromosome appears to have a single arm.

(122) Answer : (1)
Solution:
During the S-phase, chromosome number remains the same, but DNA content doubles.

(123) Answer : (3)
Solution:
Outer membrane of chloroplast is relatively more permeable and has porins.
Grana is the active site for light reaction.
Stroma is the site for dark reaction or fixation of CO_2 into sugar.

(124) Answer : (1)
Hint:
RER provides precursors of enzymes for the formation of lysosomes in Golgi complex.
Solution:
SER helps in detoxification of drugs. Lysosomes are rich in hydrolytic enzymes. Golgi-complex is the important site of formation of glycoproteins.

(125) Answer : (1)
Solution:
Golgi apparatus was first observed and described by Camillo Golgi. Camillo Golgi described Golgi apparatus as a densely stained reticular structures present near the nucleus of the cell. Therefore, these were given the name Golgi body, after his name.

(126) Answer : (4)
Solution:
By the end of telophase II, four haploid cells are formed. Reductional division is the mechanism by which conservation of the specific chromosome of each species is achieved across generations in sexually reproducing organisms.
Interkinesis is followed by prophase II.

(127) Answer : (3)
Solution:
In plants, cytokinesis occurs by formation of cell plate while in animal cells, it occurs by formation of furrow.

(128) Answer : (1)
Solution:
Cymose inflorescence show limited growth of inflorescence axis.

(129) Answer : (1)
Hint:
Maize is a monocot.
Solution:
In maize seed
1. Single cotyledon is known as scutellum
2. It is endospermic
3. Seed coat is membranous and thin
4. Radicle is enclosed in coleorhiza
5. Pericarp i.e. fruit wall fuses with seed coat.

(130) Answer : (3)
Solution:
Chemiosmosis process requires a membrane, a proton pump, a proton gradient and ATP synthase.

(131) Answer : (3)
Hint:
Neutrophils and monocytes are both phagocytic.
Solution:
Both neutrophils and monocytes are similar in action as they engulf the foreign pathogens.

(132) Answer : (4)
Solution:
Loop of Henle of juxtamedullary nephron penetrates inner zone of medulla.
In humans, roughly 85% of nephrons are cortical nephrons.

(133) Answer : (3)
Solution:
The osmolarity of glomerular filtrate in the hair pin loop is 1200 $mosmL^{-1}$.

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(134) Answer : (4)
Solution:
Answer (4)
More developed sarcoplasmic reticulum.

(135) Answer : (4)
Solution:
Answer (4)
Neurotransmitters like acetylcholine help in transmission of nerve impulses at neuromuscular junctions.

(136) Answer : (3)
Solution:
Sarcomere: Functional unit of contraction.
Muscle fibre: Anatomical unit of muscle.

(137) Answer : (1)
Solution:
Two occipital condyles, which are oval processes with convex surfaces on either side of the foramen magnum, articulate with the depression of the first cervical vertebra (called atlas) to form the joint, which allows us for nodding of head.
The odontoid process of the axis forms a pivot joint with the atlas.

(138) Answer : (3)
Solution:
In neurons, the axon is a long fibre, the distal end of which is branched. In PNS, the terminal branch of axon is called axon terminal which is unmyelinated.

(139) Answer : (3)
Solution:
Many subconscious activities of our body like heart beat, breathing, digestion, forming habits, storing memories, *etc.*, are controlled in the lower areas of the brain, *i.e.,* in the medulla oblongata, pons, mesencephalon, hypothalamus, thalamus, cerebellum, amygdala, and basal ganglia.

(140) Answer : (2)
Solution:
High blood glucose level (hyperglycemia) inhibits the release of glucagon (negative feedback).
Glucagon acts mainly on the liver cells (hepatocytes) and stimulates glycogenolysis, resulting in an increased blood sugar level.

(141) Answer : (4)
Solution:
PTH stimulates osteoclast activity and bone resorption.
Parathormone is released by the parathyroid glands, which are four small glands located on the backside of the thyroid gland.
Tetany is caused by low blood calcium level (hypocalcemia), which happens when there is insufficient PTH (hypoparathyroidism).
The follicles of the thyroid gland are lined by simple cuboidal epithelium.
PTH cause Ca^{+2} reabsorption from the renal tubule.

(142) Answer : (3)
Solution:
Annelids, arthropods and chordates exhibit metamerism.

(143) Answer : (4)
Solution:
Antennal glands are the excretory structures in Prawns.

(144) Answer : (4)
Solution:
Subphylum Vertebrata is divided into two divisions, Agnatha and Gnathostomata. Division Gnathostomata further divided into two super classes, Pisces and Tetrapoda. Hence, we can say all gnathostomes are vertebrates but all vertebrates are not gnathostomes. Similarly, all tetrapods are gnathostomes, but all gnathostomes are not tetrapods.

(145) Answer : (1)
Solution:
Rana belongs to class Amphibia.
Psittacula is a bird (4-chambered heart)
Fertilisation is internal in *Balaenoptera* (mammal)

(146) Answer : (2)
Solution:

10th segment in both the sexes of cockroach, bears a pair of anal cerci. Anal style is present only in males. Hypopharynx acts as tongue, lies within the cavity enclosed by the mouth parts. Mandibles possess grinding and incising regions on them.

(147) Answer : (1)
Solution:

HCl and gastric juice	– Secreted from the walls of stomach
Bile and pancreatic juice	– Secreted from liver and pancreas respectively
Villi and microvilli	– Finger-like folds in intestine

(148) Answer : (1)
Solution:
The entire foregut is lined by cuticle. The foregut does not involve significant absorption. The gizzard helps in grinding of food particles. Digestive juices are secreted by gastric caeca present at the junction of foregut and midgut.

(149) Answer : (1)
Solution:
In human females, oogenesis starts during embryonic development. Acrosome is filled with enzymes. A fluid-filled cavity called antrum is present in the tertiary follicle. First polar body is very small in comparison to secondary oocyte.

(150) Answer : (4)
Hint:
LH and thyrocalcitonin are secreted by permanent endocrine glands.
Solution:
In pregnancy the levels of estrogen, progesterone, cortisol, prolactin, thyroxine, *etc*., are increased several folds.
Placenta acts as a temporary endocrine tissue, produces several hormones like hCG, hPL, estrogens, progestogens, *etc*. In the later phase of pregnancy, a hormone called relaxin is also secreted by the ovaries. LH is secreted and released by pituitary.

(151) Answer : (3)
Solution:
Mitosis occurs in male germ cells (spermatogonial cells) and this occurs continuously in males. The level of testosterone decreases in later stages. It is not always secreted in a pulsatile manner.

(152) Answer : (4)
Solution:
The completion of second meiotic division of the secondary oocyte and the formation of second polar body and a haploid ovum is a part of fertilization event.

(153) Answer : (3)
Solution:
In ICSI, sperm is directly injected into the ovum.

(154) Answer : (4)
Solution:
In the mentioned case multiple mechanisms acting simultaneously.
- Suppressed LH $\rightarrow$No ovulation
- Progesterone thickens cervical mucus
- Altered endometrium prevents implantation
- FSH suppression inhibits follicular maturation.

(155) Answer : (3)
Solution:
Ernst Haeckel proposed the embryological support for evolution based upon observation of certain features during embryonic stage common to all vertebrates but absent in adult.

(156) Answer : (4)
Solution:
Dictyota is a brown alga. Rest all are red algae.

(157) Answer : (3)
Solution:
In gymnosperms and bryophytes, the male and female sex organs are not always borne on different trees.

(158) Answer : (3)
Solution:
Puccinia is an example of Basidiomycetes and it is a rust fungus. Conidia are not found in *Puccinia*.

(159) Answer : (1)
Solution:

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Sleeping sickness is caused by *Trypanosoma*.
BSE is caused by prions.

(160) Answer : (3)
Solution:
Deletions and insertions of base pairs of DNA cause frame-shift mutations.

(161) Answer : (2)
Solution:
(a) $I^A i \times I^B i$
↓
$I^A i, I^A I^B, i i, i I^B$
[4 genotypes, 4 phenotypes]
(b) $I^B I^B \times I^A i$
↓
$I^A I^B, I^B i, I^A I^B, I^B i$
[2 genotypes, 2 phenotypes]
(c) $I^A I^B \times I^B i$
↓
$I^A I^B, I^A i, I^B I^B, I^B i$
[4 genotypes, 3 phenotypes]
(d) $I^A I^A \times I^A I^B$
↓
$I^A I^A, I^A I^B, I^A I^A, I^A I^B$
[2 genotypes, 2 phenotypes]

(162) Answer : (1)
Solution:
Two nucleotide are linked through 3′ → 5′ phosphodiester linkage to form polynucleotide chain.

(163) Answer : (2)
Hint:
The DNA dependent DNA polymerase catalyses polymerisation only in one direction, that is, 5' → 3'.
Solution:
The replication is continuous on one template strand with polarity 3' → 5' and is known as leading daughter strand.

(164) Answer : (3)
Solution:
According to Chargaff's rule,
Amount of purine = Amount of pyrimidine
Thymine = 20%
Adenine = 20%
A + T + G + C = 100%
20 + 20 + G + C = 100%
G + C = 100 – 40 = 60%
G = C; G = 30%; C = 30%

(165) Answer : (4)
Hint:
Post-transcriptional modification of mRNA occurs in eukaryotes.
Solution:
In prokaryotes, the mRNA does not require any processing to become active. In eukaryotes, translation is not possible if cap (methyl guanosine triphosphate) is lacking, because cap is identified by 18S rRNA of ribosomal unit.

(166) Answer : (4)
Solution:
A-Antipodals, B-Polar nuclei, C-Synergids, D-Egg

(167) Answer : (1)

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Solution:
According to 10% rule, energy at a lower trophic level is always more than that at a higher trophic level. Pyramid of energy is always upright.

(168) Answer : (2)
Solution:
Vertical distribution of different species occupying different levels in an ecosystem is called stratification.

(169) Answer : (1)
Solution:
Abingdon tortoise in Galapagos Islands became extinct within a decade after goats were introduced on the island. The type of interaction between them is competition.

(170) Answer : (1)
Hint:
Methanogens are responsible for the production of methane and they are present in the gut of ruminant animals.
Solution:
Thermoacidophiles are found in hot water springs having low pH.

(171) Answer : (2)
Hint:
Oparin and Haldane talked about the prebiotic soup.
Solution:
Conventional religious literature tells us about the theory of special creation. This theory has three connotations. One, that all living organisms (species or types) that we see today were created as such. Two, that the diversity was always the same since creation and will be the same in future also. Three, that Earth is about 4000 years old.
Louis Pasteur by careful experimentation demonstrated that life comes from pre-existing life.
Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules and that formation of life was preceded by chemical evolution.

(172) Answer : (4)
Solution:
Morphine is obtained from the plant *Papaver somniferum*.

(173) Answer : (2)
Solution:
Several genes called cellular oncogenes or proto oncogenes have been identified in normal cells which when activated under certain conditions, could lead to oncogenic transformation of cells.

(174) Answer : (1)
Solution:
IgA is dimeric and IgM is pentameric. IgG have the ability to cross the placental barrier.

(175) Answer : (4)
Solution:
Since, we are performing restriction digestion only with *Sma I* and *Pvu II,* we will obtain 3 DNA fragments. The obtained DNA fragments will be of the sizes 7kb, 2kb and 2kb. The DNA fragments of the sizes 2kb will travel equal distances on the gel, thus, a single thick band will be seen in the gel image; and the 7 kb DNA fragment will be present close to the cathode.

(176) Answer : (3)
Solution:
A stirred-tank reactor is usually cylindrical or with a curved base to facilitate the mixing of the reactor's contents.
In a continuous culture system, the used medium is drained out from one side while fresh medium is added from the other to maintain the cells in their physiologically most active log/exponential phase.
Aerobic fermentation requires oxygen for the growth of microorganisms and product formation. Aeration and agitation are used to maintain oxygen levels.

(177) Answer : (1)
Solution:
Hind II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs.
Transformation is a procedure through which a piece of DNA is introduced in a host bacterium.
Bioprocess engineering is the maintenance of sterile (microbial contamination-free) ambience in chemical engineering processes to enable growth of only the desired microbe/eukaryotic cell in large quantities for the manufacture of biotechnological products like antibiotics, vaccines, enzymes, *etc.*

(178) Answer : (3)
Hint:
No digestion
Solution:
A restriction enzyme cuts DNA at its specific recognition sequence only.

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The plasmid was tried to cut by using a restriction endonuclease that did not have a recognition site in the plasmid. Hence, the enzyme would not be able to cut the plasmid and the plasmid would remain undigested.

(179) Answer : (4)

Solution:

The mature insulin hormone lacks C peptide.

(180) Answer : (1)

Hint:

Polygalacturonase activity is altered in Flavr Savr tomatoes.

Solution:

Early exhaustion of soil fertility is prevented in GMO. Gene *cryIAb* controls infection by insects in corn plants.

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